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Bucket architecture for an image capture device including a reconfigurable PCB module

Abstract

An image capture device is disclosed that includes: a front housing portion; a rear housing portion including a heatsink that is connected to the front housing portion; and a printed circuit board (PCB) module that is operatively connected to the front housing portion. The PCB module includes: a front PCB assembly; a rear PCB assembly; and a connector that extends between the front PCB assembly and the rear PCB assembly. The connector includes a flexible construction that facilitates reconfiguration of the PCB module during assembly of the image capture device between a first configuration, in which the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation, and a second configuration, in which the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to an image capture device and, more specifically, to a bucket architecture for a front housing portion of the image capture device that facilitates the insertion of various internal components in a single (e.g., forward, front) direction and connection of the various internal components to the front housing portion.

BACKGROUND

(2) Image capture devices are used in a variety of applications, including, for example, handheld cameras and video recorders, cell phones, drones, etc. Image capture devices typically include one or more optical elements (e.g., lenses) that capture content by receiving and focusing light as well as one or more image sensors that convert the captured content into an electronic image signal that is processed by an image signal processor to form an image. In some image capture devices, the optical element(s) and the image sensor(s) are combined into a single unit, which is known as an integrated sensor-lens assembly (ISLA). In addition to the ISLA, image capture devices include an array of internal components, both structural and electrical, that require precise placement and alignment, which results in a cumbersome and complex assembly process. As such, an opportunity exists to improve not only the assembly of image capture devices, but product packaging.

(3) The present disclosure addresses this opportunity by providing a bucket architecture for a front housing portion of an image capture device that simplifies assembly and contributes to a reduction in the overall form factor.

SUMMARY

(4) In one aspect of the present disclosure, an image capture device is disclosed that includes: a front housing portion that defines an internal compartment; a rear housing portion that is connected to the front housing portion so as to form a watertight seal therebetween; a tray that is positioned between the front housing portion and the rear housing portion; a power source that is supported by the tray; a heatsink that is connected to the front housing portion and which is configured to distribute thermal energy through the image capture device; an integrated sensor-lens assembly (ISLA) that extends through the heatsink and which includes an image sensor and a lens that is positioned to receive and direct light onto the image sensor; a mounting member that is connected to the front housing portion and to the ISLA such that the ISLA is indirectly connected to the front housing portion via the mounting member; a printed circuit board (PCB) module that is connected to the heatsink; and a chassis.

(5) The PCB module includes: a PCB assembly; a rear PCB assembly; and a flexible connector that extends between the front and rear PCB assemblies. The front and rear PCB assemblies each support one or more electrical and/or thermal components of the image capture device, and the flexible connector facilitates reconfiguration of the PCB module between a first configuration and a second configuration during assembly of the image capture device. In the first configuration, the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation, and in the second configuration, the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation.

(6) The chassis is positioned between the front PCB assembly and the rear PCB assembly and includes a frame that structurally supports the PCB module.

(7) In certain embodiments, the front housing portion may define a window that is configured to receive the mounting member such that the mounting member extends through the window and into the internal compartment.

(8) In certain embodiments, the mounting member may include a front end portion that extends externally of the image capture device and a rear end portion that extends into the image capture

device.
(9) In certain embodiments, the ISLA may be connected to the rear end portion of the mounting member.
(10) In certain embodiments, the front end portion of the mounting member may be configured for releasable connection to a cover for the image capture device such that the cover is connectable to and disconnectable from the image capture device via the mounting member.
(11) In certain embodiments, the tray may define through-bores that are configured to receive mounts extending inwardly from the front housing portion.
(12) In certain embodiments, the mounts may be configured to receive fasteners that connect the front housing portion to an interconnect mechanism.
(13) In certain embodiments, the interconnect mechanism may be configured for engagement with an accessory such that the image capture device is connectable to the accessory via the interconnect mechanism.
(14) In certain embodiments, the tray may include a front flange that engages the front housing portion and a rear flange that engages the rear housing portion.
(15) In certain embodiments, the PCB module may be generally L-shaped in the first configuration and generally U-shaped in the second configuration.
(16) In certain embodiments, the PCB module may define a generally U-shaped chamber in the second configuration that is configured to receive the chassis such that the chassis is positioned between the front PCB assembly and the rear PCB assembly.
(17) In certain embodiments, the front PCB assembly and the rear PCB assembly may each include a rigid body with a laminated construction defined by layers that are arranged in adjacent relation.
(18) In certain embodiments, the flexible connector may include a first end portion and a second end portion.
(19) In certain embodiments, the first end portion may be positioned within the rigid body of the front PCB assembly between the layers thereof, and the second end portion may be positioned within the rigid body of the rear PCB assembly between the layers thereof.
(20) In certain embodiments, the chassis may include at least one electrical connector (e.g., a USB connector).
(21) In another aspect of the present disclosure, an image capture device is disclosed that includes: a front housing portion; a rear housing portion including a heatsink that is connected to the front housing portion; and a printed circuit board (PCB) module that is operatively connected to the front housing portion. The PCB module includes: a front PCB assembly; a rear PCB assembly; and a connector that extends between the front PCB assembly and the rear PCB assembly. The connector includes a flexible construction that facilitates reconfiguration of the PCB module during assembly of the image capture device between a first configuration, in which the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation, and a second configuration, in which the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation.
(22) In certain embodiments, the rear PCB assembly may be thermally connected to the rear housing portion to facilitate thermal energy transfer from the rear PCB assembly to the heatsink.
(23) In certain embodiments, the rear PCB assembly may include a system-on-chip and at least one power management integrated circuit.
(24) In certain embodiments, the connector may include a first end portion that extends into the front PCB assembly and a second end portion that extends into the rear PCB assembly.
(25) In certain embodiments, the front PCB assembly may be laminated in construction and may include layers that are positioned about the first end portion of the connector, and the rear PCB assembly may be laminated in construction and may include layers that are positioned about the second end portion of the connector.
(26) In another aspect of the present disclosure, a method of assembling an image capture device is disclosed. The method includes: connecting a first heatsink to a front housing portion of the image

capture device; inserting a printed circuit board (PCB) module including a front PCB assembly and a rear PCB assembly into the front housing portion with the PCB module in a first configuration in which the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation; reconfiguring the PCB module from the first configuration into a second configuration in which the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation; and connecting the PCB module to the first heatsink.
(27) In certain embodiments, the method may further include thermally connecting the PCB module to a rear housing portion including a second heatsink, wherein the first heatsink and the second heatsink are formed as discrete components of the image capture device.
(28) In certain embodiments, thermally connecting the PCB module to the rear housing portion may include thermally connecting the second heatsink to a system-on-chip and at least one power management integrated circuit.
(29) In certain embodiments, reconfiguring the PCB module from the first configuration into the second configuration may include orienting at least one first component on the PCB module such that that the at least one first component is directed outwardly (e.g., towards the first heatsink and the second heatsink) and orienting at least one second component on the PCB module such that the at least one second component is directed inwardly (e.g., away from the first heatsink and the second heatsink).
(30) In certain embodiments, the at least one first component may include a system-on-chip and at least one power management integrated circuit, and the at least one second component may include at least one integrated circuit.
(31) In certain embodiments, the method may further include thermally connecting the at least one first component to the first heatsink and the second heatsink.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. According to common practice, the various features of the drawings may not be to-scale, and the dimensions of the various features may be arbitrarily expanded or reduced. Additionally, in the interest of clarity, certain components, elements, and/or features may be omitted from certain drawings in the interest of clarity.
(2) FIGS. **1**A-B are isometric views of an example of an image capture device.
(3) FIGS. **2**A-B are isometric views of another example of an image capture device.
(4) FIG. **3** is a block diagram of electronic components of an image capture device.
(5) FIG. **4** is a front, perspective view of another example of an image capture device.
(6) FIG. **5** is a rear, perspective view of the image capture device seen in FIG. **4**.
(7) FIG. **6** is a rear, perspective view of a front housing portion of the image capture device seen in FIG. **4**.
(8) FIG. **7** is a rear, perspective view of a rear housing portion of the image capture device seen in FIG. **4**.
(9) FIG. **8** is a front, perspective view of the rear housing portion seen in FIG. **7**.
(10) FIG. **9** is a rear, perspective view of the of the image capture device seen in FIG. **4** with the rear housing portion removed and illustrating a power source, a tray supporting the power source, and a PCB module.
(11) FIG. **10** is a rear, perspective view of the image capture device seen in FIG. **9** with the power source, the tray, and the PCB module removed and illustrating an integrated sensor-lens assembly (ISLA) shown connected to a mounting member.
(12) FIG. **11** is a front, perspective view of the image capture device seen in FIG. **4** with parts

separated.
(13) FIG. **12** is a rear, perspective view of the image capture device seen in FIG. **4** with parts separated.
(14) FIG. **13** is a partial, vertical, cross-sectional view of the image capture device seen in FIG. **4** illustrating engagement between the front housing portion and the tray.
(15) FIG. **14** is a partial, vertical, cross-sectional view of the image capture device seen in FIG. **4** illustrating engagement between the rear housing portion and the tray.
(16) FIGS. **15** and **16** are respective front and rear perspective views of the image capture device seen in FIG. **4** illustrating a variety of sealing members.
(17) FIG. **17** is a rear, perspective view of the image capture device seen in FIG. **4** with the rear housing portion removed and illustrating the PCB module shown with a chassis.
(18) FIGS. **18** and **19** are rear, perspective views of the PCB module and the chassis seen in FIG. **17** shown connected to a heatsink.
(19) FIG. **20** is a side, plan view of the PCB module, the chassis, and the heatsink seen in FIGS. **18** and **19**.
(20) FIG. **21** is a top, plan view of the PCB module, the chassis, and the heatsink seen in FIGS. **18-20**.
(21) FIG. **22** is a rear, perspective view of the PCB module shown in a first configuration.
(22) FIG. **23** is a rear, perspective view of the PCB module shown in a second configuration.
DETAILED DESCRIPTION
(23) The present disclosure describes an image capture device including a bucket architecture that contributes to a reduction in the overall form factor of the image capture device and simplifies the assembly process. The reduced form factor of the image capture device facilitated by the bucket architecture described herein not only allows for improved mounting schemes, but supports mounting to an increased variety of accessories. For example, the reduced form factor increases user comfort when the image capture device is used with a wearable accessory (e.g., a helmet, etc.), as compared to known image capture devices, which often create a more awkward, uncomfortable, and cumbersome experience due to their larger size and increased weight.
(24) The image capture device described herein includes a rear housing portion and a front housing portion (e.g., a bucket) that is configured to receive the various internal components of the image capture device (e.g., the heatsink, the ISLA, etc.) in a single (e.g., forward, front) direction of insertion. To facilitate such insertion and connection of the various internal components to the front housing portion in the manner described herein, the image capture device includes a reconfigurable PCB module having front and rear PCB assemblies and a (flexible) connector that extends therebetween. The flexible construction of the connector allows for deformation thereof (e.g., bending), which facilitates reconfiguration of the PCB module between first and second configurations during assembly of the image capture device. In the first configuration, the front and rear PCB assemblies are oriented in non-parallel relation, and in the second configuration, the front and rear PCB assemblies are oriented in generally parallel relation. More specifically, in the first configuration, the PCB module includes a generally L-shaped configuration, whereas in the second configuration, the PCB module includes a generally U-shaped configuration in which the front and rear PCB assemblies collectively define a (generally U-shaped) chamber therebetween. The generally U-shaped second configuration of the PCB module allows various electrical and/or thermal components of the image capture device to be included on a single PCB module. The incorporation of a single, reconfigurable PCB module allows for a more compact internal design and, thus, further reductions in the overall form factor of the image capture device, and simplifies the overall assembly process by improving not only access to the internal component(s) (e.g., those housed (accommodated) between the front and rear PCB assemblies), but visibility of the internal component(s) as well by reducing (if not entirely eliminating) blind steps.
(25) FIGS. **1**A-B are isometric views of an example of an image capture device **100**. The image

capture device **100** may include a body **102**, one or more lenses **104** structured on a front surface of the body **102**, various indicators on the front surface of the body **102** (such as light-emitting diodes (LEDs), displays, and the like), various input mechanisms (such as buttons, switches, and/or touch-screens), and electronics (such as imaging electronics, power electronics, etc.) internal to the body **102** for capturing images via the lens(es) **104** and/or performing other functions. The lens(es) **104** receive light incident upon the lens(es) **104** and to direct received light onto an image sensor internal to the body **102**, as described in further detail below. The image capture device **100** may be configured to capture images and video and to store captured images and video for subsequent display or playback.

(26) The image capture device **100** may include an LED or another form of indicator **106** to indicate a status of the image capture device **100** and a liquid-crystal display (LCD) or other form of a display **108** to show status information such as battery life, camera mode, elapsed time, and the like. The image capture device **100** may also include a mode button **110** and a shutter button **112** that are configured to allow a user of the image capture device **100** to interact with the image capture device **100**. For example, the mode button **110** and the shutter button **112** may be used to turn the image capture device **100** on and off, scroll through modes and settings, and select modes and change settings. The image capture device **100** may include additional buttons or interfaces (not shown) to support and/or control additional functionality.

(27) The image capture device **100** may include a door **114** coupled to the body **102**, for example, using a hinge mechanism **116**. The door **114** may be secured to the body **102** using a latch mechanism **118** that releasably engages the body **102** at a position generally opposite the hinge mechanism **116**. The door **114** may also include a seal **120** and a battery interface **122**. When the door **114** is an open position, access is provided to an input-output (I/O) interface **124** for connecting to or communicating with external devices as described below and to a battery receptacle **126** for placement and replacement of a battery (not shown). The battery receptacle **126** includes operative connections (not shown) for power transfer between the battery and the image capture device **100**. When the door **114** is in a closed position, the seal **120** engages a flange (not shown) or other interface to provide an environmental seal, and the battery interface **122** engages the battery to secure the battery in the battery receptacle **126**. The door **114** can also have a removed position (not shown) where the entire door **114** is separated from the image capture device **100**, that is, where both the hinge mechanism **116** and the latch mechanism **118** are decoupled from the body **102** to allow the door **114** to be removed from the image capture device **100**.

(28) The image capture device **100** may include a microphone **128** on a front surface and another microphone **130** on a side surface. The image capture device **100** may include other microphones on other surfaces (not shown). The microphones **128**, **130** may be configured to receive and record audio signals in conjunction with recording video or separate from recording of video. The image capture device **100** may include a speaker **132** on a bottom surface of the image capture device **100**. The image capture device **100** may include other speakers on other surfaces (not shown). The speaker **132** may be configured to play back recorded audio or emit sounds associated with notifications.

(29) A front surface of the image capture device **100** may include a drainage channel **134**. A bottom surface of the image capture device **100** may include a (first) interconnect mechanism **136** that is configured for engagement with (connecting to) an accessory, handle, grip, etc., such that the image capture device **100** is (repeatably) connectable to the accessory via the interconnect mechanism **136**. In the example shown in FIG. **1**B, the interconnect mechanism **136** includes folding protrusions **137** (e.g., fingers **137**A) that are configured to move between a nested (collapsed) position (FIG. **1**B) and an extended (open) position (FIGS. **2**A, **2**B) that facilitates coupling of the protrusions **137** to mating protrusions of other devices such as accessories, handle grips, mounts, clips, or like devices, as discussed in further detail below. More specifically, the interconnect mechanism **136** includes a (first) protrusion **137***i* defining a (first) opening **137**Bi and a (second)

protrusion **137***ii* defining a (second) opening **137**Bii, which are moveable between the nested and extended positions independently of each other. As seen in FIG. **1**B, for example, when the protrusions **137** are in the nested position, the protrusions **137** are received within (accommodated by) corresponding cavities **103**, which extend (vertically upward) into the body **102** (e.g., towards the shutter button **112**), and when the protrusions **137** are in the extended position, the protrusions **137** are removed from the cavities **103** such that the protrusions **137** extend (vertically downward) from the body (e.g., away from the shutter button **112**). More specifically, the body **102** includes a (first) cavity **103***i* that is configured to receive the protrusion **137***i* when the protrusion **137***i* is in the nested position and a (second) cavity **103***ii* that is configured to receive the protrusion **137***ii* when the protrusion **137***ii* is in the nested position. To facilitate reception of the protrusions **137**, the cavities **103** include identical (or generally identical) configurations, which correspond to those defined by the protrusions **137**. As such, in the illustrated embodiment, the cavities **103** each include a D-shaped (or generally D-shaped) transverse (e.g., horizontal) cross-sectional configuration. It should be appreciated, however, that the particular configurations of the protrusions **137** and the cavities **103** may be altered in various embodiments without departing from the scope of the present disclosure.
(30) The image capture device **100** may include an interactive display **138** that allows for interaction with the image capture device **100** while simultaneously displaying information on a surface of the image capture device **100**.
(31) The image capture device **100** of FIGS. **1**A-B includes an exterior that encompasses and protects internal electronics. In the present example, the exterior includes six surfaces (i.e., a front face, a left face, a right face, a back face, a top face, and a bottom face) that form a rectangular cuboid. Furthermore, both the front and rear surfaces of the image capture device **100** are rectangular. In other embodiments, the exterior may have a different shape. The image capture device **100** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass. The image capture device **100** may include features other than those described here. For example, the image capture device **100** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **100**.
(32) The image capture device **100** may include various types of image sensors, such as charge-coupled device (CCD) sensors, active pixel sensors (APS), complementary metal-oxide-semiconductor (CMOS) sensors, N-type metal-oxide-semiconductor (NMOS) sensors, and/or any other image sensor or combination of image sensors.
(33) Although not illustrated, in various embodiments, the image capture device **100** may include other additional electrical components (e.g., an image processor, camera system-on-chip (SoC), etc.), which may be included on one or more circuit boards within the body **102** of the image capture device **100**.
(34) The image capture device **100** may interface with or communicate with an external device, such as an external user interface device (not shown), via a wired or wireless computing communication link (e.g., the I/O interface **124**). Any number of computing communication links may be used. The computing communication link may be a direct computing communication link or an indirect computing communication link, such as a link including another device or a network, such as the internet, may be used.
(35) In some implementations, the computing communication link may be a Wi-Fi link, an infrared link, a Bluetooth (BT) link, a cellular link, a ZigBee link, a near field communications (NFC) link, such as an ISO/IEC 20643 protocol link, an Advanced Network Technology interoperability (ANT+) link, and/or any other wireless communications link or combination of links.
(36) In some implementations, the computing communication link may be an HDMI link, a USB link, a digital video interface link, a display port interface link, such as a Video Electronics Standards Association (VESA) digital display interface link, an Ethernet link, a Thunderbolt link,

and/or other wired computing communication link.

(37) The image capture device **100** may transmit images, such as panoramic images, or portions thereof, to the external user interface device via the computing communication link, and the external user interface device may store, process, display, or a combination thereof the panoramic images.

(38) The external user interface device may be a computing device, such as a smartphone, a tablet computer, a phablet, a smart watch, a portable computer, personal computing device, and/or another device or combination of devices configured to receive user input, communicate information with the image capture device **100** via the computing communication link, or receive user input and communicate information with the image capture device **100** via the computing communication link.

(39) The external user interface device may display, or otherwise present, content, such as images or video, acquired by the image capture device **100**. For example, a display of the external user interface device may be a viewport into the three-dimensional space represented by the panoramic images or video captured or created by the image capture device **100**.

(40) The external user interface device may communicate information, such as metadata, to the image capture device **100**. For example, the external user interface device may send orientation information of the external user interface device with respect to a defined coordinate system to the image capture device **100**, such that the image capture device **100** may determine an orientation of the external user interface device relative to the image capture device **100**.

(41) Based on the determined orientation, the image capture device **100** may identify a portion of the panoramic images or video captured by the image capture device **100** for the image capture device **100** to send to the external user interface device for presentation as the viewport. In some implementations, based on the determined orientation, the image capture device **100** may determine the location of the external user interface device and/or the dimensions for viewing of a portion of the panoramic images or video.

(42) The external user interface device may implement or execute one or more applications to manage or control the image capture device **100**. For example, the external user interface device may include an application for controlling camera configuration, video acquisition, video display, or any other configurable or controllable aspect of the image capture device **100**.

(43) The user interface device, such as via an application, may generate and share, such as via a cloud-based or social media service, one or more images, or short video clips, such as in response to user input. In some implementations, the external user interface device, such as via an application, may remotely control the image capture device **100** such as in response to user input.

(44) The external user interface device, such as via an application, may display unprocessed or minimally processed images or video captured by the image capture device **100** contemporaneously with capturing the images or video by the image capture device **100**, such as for shot framing or live preview, and which may be performed in response to user input. In some implementations, the external user interface device, such as via an application, may mark one or more key moments contemporaneously with capturing the images or video by the image capture device **100**, such as with a tag or highlight in response to a user input or user gesture.

(45) The external user interface device, such as via an application, may display or otherwise present marks or tags associated with images or video, such as in response to user input. For example, marks may be presented in a camera roll application for location review and/or playback of video highlights.

(46) The external user interface device, such as via an application, may wirelessly control camera software, hardware, or both. For example, the external user interface device may include a web-based graphical interface accessible by a user for selecting a live or previously recorded video stream from the image capture device **100** for display on the external user interface device.

(47) The external user interface device may receive information indicating a user setting, such as an

image resolution setting (e.g., 3840 pixels by 2160 pixels), a frame rate setting (e.g., 60 frames per second (fps)), a location setting, and/or a context setting, which may indicate an activity, such as mountain biking, in response to user input, and may communicate the settings, or related information, to the image capture device **100**.

(48) FIGS. **2**A-B illustrate another example of an image capture device **200**. The image capture device **200** includes a body **202** and two camera lenses **204** and **206** disposed on opposing surfaces of the body **202**, for example, in a back-to-back configuration, Janus configuration, or offset Janus configuration. The body **202** of the image capture device **200** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass.

(49) The image capture device **200** includes various indicators on the front of the surface of the body **202** (such as LEDs, displays, and the like), various input mechanisms (such as buttons, switches, and touch-screen mechanisms), and electronics (e.g., imaging electronics, power electronics, etc.) internal to the body **202** that are configured to support image capture via the two camera lenses **204** and **206** and/or perform other imaging functions.

(50) The image capture device **200** includes various indicators, for example, LEDs **208**, **210** to indicate a status of the image capture device **100**. The image capture device **200** may include a mode button **212** and a shutter button **214** configured to allow a user of the image capture device **200** to interact with the image capture device **200**, to turn the image capture device **200** on, and to otherwise configure the operating mode of the image capture device **200**. It should be appreciated, however, that, in alternate embodiments, the image capture device **200** may include additional buttons or inputs to support and/or control additional functionality.

(51) The image capture device **200** may include an interconnect mechanism **216** for connecting the image capture device **200** to a handle grip or other securing device. In the example shown in FIGS. **2**A and **2**B, the interconnect mechanism **216** includes folding protrusions configured to move between a nested or collapsed position (not shown) and an extended or open position as shown that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.

(52) The image capture device **200** may include audio components **218**, **220**, **222** such as microphones configured to receive and record audio signals (e.g., voice or other audio commands) in conjunction with recording video. The audio component **218**, **220**, **222** can also be configured to play back audio signals or provide notifications or alerts, for example, using speakers. Placement of the audio components **218**, **220**, **222** may be on one or more of several surfaces of the image capture device **200**. In the example of FIGS. **2**A and **2**B, the image capture device **200** includes three audio components **218**, **220**, **222**, with the audio component **218** on a front surface, the audio component **220** on a side surface, and the audio component **222** on a back surface of the image capture device **200**. Other numbers and configurations for the audio components are also possible.

(53) The image capture device **200** may include an interactive display **224** that allows for interaction with the image capture device **200** while simultaneously displaying information on a surface of the image capture device **200**. The interactive display **224** may include an I/O interface, receive touch inputs, display image information during video capture, and/or provide status information to a user. The status information provided by the interactive display **224** may include battery power level, memory card capacity, time elapsed for a recorded video, etc.

(54) The image capture device **200** may include a release mechanism **225** that receives a user input to in order to change a position of a door (not shown) of the image capture device **200**. The release mechanism **225** may be used to open the door (not shown) in order to access a battery, a battery receptacle, an I/O interface, a memory card interface, etc. (not shown) that are similar to components described in respect to the image capture device **100** of FIGS. **1**A and **1**B.

(55) In some embodiments, the image capture device **200** described herein includes features other than those described. For example, instead of the I/O interface and the interactive display **224**, the image capture device **200** may include additional interfaces or different interface features. For

example, the image capture device **200** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **200**.
(56) FIG. **3** is a block diagram of electronic components in an image capture device **300**. The image capture device **300** may be a single-lens image capture device, a multi-lens image capture device, or variations thereof, including an image capture device with multiple capabilities such as use of interchangeable integrated sensor lens assemblies. The description of the image capture device **300** is also applicable to the image capture devices **100**, **200** of FIGS. **1**A-B and **2**A-D.
(57) The image capture device **300** includes a body **302** which includes electronic components such as capture components **310**, a processing apparatus **320**, data interface components **330**, movement sensors **340**, power components **350**, and/or user interface components **360**.
(58) The capture components **310** include one or more image sensors **312** for capturing images and one or more microphones **314** for capturing audio.
(59) The image sensor(s) **312** is configured to detect light of a certain spectrum (e.g., the visible spectrum or the infrared spectrum) and convey information constituting an image as electrical signals (e.g., analog or digital signals). The image sensor(s) **312** detects light incident through a lens coupled or connected to the body **302**. The image sensor(s) **312** may be any suitable type of image sensor, such as a charge-coupled device (CCD) sensor, active pixel sensor (APS), complementary metal-oxide-semiconductor (CMOS) sensor, N-type metal-oxide-semiconductor (NMOS) sensor, and/or any other image sensor or combination of image sensors. Image signals from the image sensor(s) **312** may be passed to other electronic components of the image capture device **300** via a bus **380**, such as to the processing apparatus **320**. In some implementations, the image sensor(s) **312** includes a digital-to-analog converter. A multi-lens variation of the image capture device **300** can include multiple image sensors **312**.
(60) The microphone(s) **314** is configured to detect sound, which may be recorded in conjunction with capturing images to form a video. The microphone(s) **314** may also detect sound in order to receive audible commands to control the image capture device **300**.
(61) The processing apparatus **320** may be configured to perform image signal processing (e.g., filtering, tone mapping, stitching, and/or encoding) to generate output images based on image data from the image sensor(s) **312**. The processing apparatus **320** may include one or more processors having single or multiple processing cores. In some implementations, the processing apparatus **320** may include an application specific integrated circuit (ASIC). For example, the processing apparatus **320** may include a custom image signal processor. The processing apparatus **320** may exchange data (e.g., image data) with other components of the image capture device **300**, such as the image sensor(s) **312**, via the bus **380**.
(62) The processing apparatus **320** may include memory, such as a random-access memory (RAM) device, flash memory, or another suitable type of storage device, such as a non-transitory computer-readable memory. The memory of the processing apparatus **320** may include executable instructions and data that can be accessed by one or more processors of the processing apparatus **320**. For example, the processing apparatus **320** may include one or more dynamic random-access memory (DRAM) modules, such as double data rate synchronous dynamic random-access memory (DDR SDRAM). In some implementations, the processing apparatus **320** may include a digital signal processor (DSP). More than one processing apparatus may also be present or associated with the image capture device **300**.
(63) The data interface components **330** enable communication between the image capture device **300** and other electronic devices, such as a remote control, a smartphone, a tablet computer, a laptop computer, a desktop computer, or a storage device. For example, the data interface components **330** may be used to receive commands to operate the image capture device **300**, transfer image data to other electronic devices, and/or transfer other signals or information to and from the image capture device **300**. The data interface components **330** may be configured for

wired and/or wireless communication. For example, the data interface components **330** may include an I/O interface **332** that provides wired communication for the image capture device, which may be a USB interface (e.g., USB type-C), a high-definition multimedia interface (HDMI), or a FireWire interface. The data interface components **330** may include a wireless data interface **334** that provides wireless communication for the image capture device **300**, such as a Bluetooth interface, a ZigBee interface, and/or a Wi-Fi interface. The data interface components **330** may include a storage interface **336**, such as a memory card slot configured to receive and operatively couple to a storage device (e.g., a memory card) for data transfer with the image capture device **300** (e.g., for storing captured images and/or recorded audio and video).

(64) The movement sensors **340** may detect the position and movement of the image capture device **300**. The movement sensors **340** may include a position sensor **342**, an accelerometer **344**, or a gyroscope **346**. The position sensor **342**, such as a global positioning system (GPS) sensor, is used to determine a position of the image capture device **300**. The accelerometer **344**, such as a three-axis accelerometer, measures linear motion (e.g., linear acceleration) of the image capture device **300**. The gyroscope **346**, such as a three-axis gyroscope, measures rotational motion (e.g., rate of rotation) of the image capture device **300**. Other types of movement sensors **340** may also be present or associated with the image capture device **300**.

(65) The power components **350** may receive, store, and/or provide power for operating the image capture device **300**. The power components **350** may include a battery interface **352** and a battery **354**. The battery interface **352** operatively couples to the battery **354**, for example, with conductive contacts to transfer power from the battery **354** to the other electronic components of the image capture device **300**. The power components **350** may also include an external interface **356**, and the power components **350** may, via the external interface **356**, receive power from an external source, such as a wall plug or external battery, for operating the image capture device **300** and/or charging the battery **354** of the image capture device **300**. In some implementations, the external interface **356** may be the I/O interface **332**. In such an implementation, the I/O interface **332** may enable the power components **350** to receive power from an external source over a wired data interface component (e.g., a USB type-C cable).

(66) The user interface components **360** may allow the user to interact with the image capture device **300**, for example, providing outputs to the user and receiving inputs from the user. The user interface components **360** may include visual output components **362** to visually communicate information and/or present captured images to the user. The visual output components **362** may include one or more lights **364** and/or more displays **366**. The display(s) **366** may be configured as a touch screen that receives inputs from the user. The user interface components **360** may also include one or more speakers **368**. The speaker(s) **368** can function as an audio output component that audibly communicates information and/or presents recorded audio to the user. The user interface components **360** may also include one or more physical input interfaces **370** that are physically manipulated by the user to provide input to the image capture device **300**. The physical input interfaces **370** may, for example, be configured as buttons, toggles, or switches. The user interface components **360** may also be considered to include the microphone(s) **314**, as indicated in dotted line, and the microphone(s) **314** may function to receive audio inputs from the user, such as voice commands.

(67) Referring now to FIGS. **4-12**, an image capture device **400** is illustrated, which includes features similar to the aforedescribed image capture devices **100**, **200**, **300** and, accordingly, will only be discussed with respect to differences therefrom in the interest of brevity. The image capture device **400** includes: a front housing portion **402** (e.g., a bucket **404**); a rear housing portion **406** that is connected to the front housing portion **402** so as to form a watertight seal therebetween; a (removable) cover **500**; and a variety of internal components, both structural and electrical. As described in detail below, the internal components of the image capture device **400** include: a tray **600** that is positioned between the housing portions **402**, **406**; a power source **700** (e.g., the

aforementioned battery **354** (FIG. **3**)); a mounting member **800** (e.g., a bayonet **802**); a (first) sealing member **900**; an integrated sensor-lens assembly (ISLA) **1000**; a (second) sealing member **1100**; a (first, front) heatsink **1200**, which is supported by (e.g., connected to) the front housing portion **402** and is configured to distribute thermal energy through the image capture device **400**; a reconfigurable printed circuit board (PCB) module **1300**; and a chassis **1400**.

(68) More specifically, FIG. **4** provides a front, perspective view of the image capture device **400**; FIG. **5** provides a rear, perspective view of the image capture device **400**; FIG. **6** provides a rear, perspective view of the front housing portion **402**; FIG. **7** provides a rear, perspective view of the rear housing portion **406**; FIG. **8** provides a front, perspective view of the rear housing portion **406**; FIG. **9** provides a rear, perspective view of the tray **600**, the power source **700**, and the PCB module **1300** upon assembly of the image capture device **400**; FIG. **10** provides a rear, perspective view of the ISLA **1000** shown connected to the mounting member **800**; FIG. **11** provides a front, perspective view of the image capture device **400** with parts separated; and FIG. **12** provides a rear, perspective view of the image capture device **400** with parts separated.

(69) The front housing portion **402** (e.g., the bucket **404**) includes an architecture that provides a framework for various components of the image capture device **400** including, for example, the tray **600**, the power source **700**, the mounting member **800**, the ISLA **1000**, the heatsink **1200**, which itself provides a framework for various components, electronics, and circuitry that support operability and functionality of the image capture device **400**, the PCB module **1300**, and the chassis **1400**. As described in detail below, the framework provided by the architecture of the front housing portion **402** simplifies assembly of the image capture device **400** by focusing the area where connections are to be made and reducing (if not entirely eliminating) blind steps, which improves visibility and facilitates more precise alignment of the components.

(70) The front housing portion **402** defines an internal compartment **408** (FIG. **6**) that receives (accommodates) the internal components of the image capture device **400** and includes a plurality (series) of apertures **410** (FIG. **4**), which are configured to receive (accommodate) various buttons, displays, etc., and a window **412**. In the particular embodiment illustrated, the front housing portion **402** includes: a (first) aperture **410***i* that is configured to receive a (first) button **414***i* (e.g., a shutter button); a (second) aperture **410***ii* that is configured to receive a (second) button **414***ii* (e.g., a mode button); a (third) aperture **410***iii* that is configured to receive a display **416**; and a (fourth) aperture **410***iv* that is configured to receive a light pipe **418** (e.g., a status indicator or the like). It should be appreciated, however, that the particular number of apertures **410**, buttons **414**, displays **416**, and light pipes **418** may be altered in various embodiments without departing from the scope of the present disclosure. As such, embodiments of the image capture device **400** including both greater and fewer numbers of apertures **410**, buttons **414**, displays **416**, and light pipes **418** are also contemplated herein and would not be beyond the scope of the present disclosure.

(71) The window **412** is configured to receive the mounting member **800**, the sealing member **900**, and the ISLA **1000** and is defined by a flange **420** (FIG. **11**). The flange **420** is configured in correspondence with the mounting member **800** (e.g., such that the flange **420** and the mounting member **800** include configurations that mirror each other), whereby the mounting member **800** seats within the window **412**, as described in further detail below. As seen in FIG. **11**, the flange **420** includes: a plurality (series) of apertures **422**; a plurality (series) of reliefs **424** (e.g., cutouts, notches, indentations, etc.); and a channel **426** that is configured to receive the sealing member **900**.

(72) The apertures **422** are configured to receive a plurality (series) of mechanical fasteners **428** (e.g., screws, pins, rivets, etc.) such that the mounting member **800** is directly connectable to the front housing portion **402** in either a fixed or removable fashion. Although shown as including four apertures **422** in the embodiment illustrated, it should be appreciated that the particular number of apertures **422** and, thus, the particular number of mechanical fasteners **428**, may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular

configuration of the mounting member **800**).

(73) The reliefs **424** are configured to receive corresponding structures on the mounting member **800**, as described in further detail below, such that the mounting member **800** is extendable through the window **412** and into the internal compartment **408**. Although shown as including three reliefs **424** in the embodiment illustrated, it should be appreciated that the particular number of reliefs **424** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the mounting member **800**).

(74) Extending around (e.g., adjacent to) its rear perimeter, the front housing portion **402** defines a planar shelf **430** (FIG. **6**) and a tongue **432** that projects rearwardly (i.e., toward the rear housing portion **406**). The planar shelf **430** defines a seat for the rear housing portion **406** to facilitate proper location (alignment) of the rear housing portion **406** relative to the front housing portion **402**, which further enhances the overall fit and finish of the image capture device **400** upon assembly. As described in further detail below, the tongue **432** is configured to interface with a corresponding structure on the rear housing portion **406** to facilitate (sealed) engagement of the housing portions **402**, **406** upon assembly of the image capture device **400**.

(75) The front housing portion **402** further includes at least one internal guide **434** (FIG. **6**) that facilitates the insertion and connection of the various internal components of the image capture device **400**. For example, in the particular embodiment illustrated, the front housing portion **402** includes a locating feature **436** that facilitates alignment of the WiFi antenna (not shown) as well as a plurality (series) of mounts **438** that are configured to support connection of the mounting member **800**, the heatsink **1200**, the tray **600**, etc., to the front housing portion **402** and facilitate proper location thereof within the internal compartment **408**. The internal guide(s) **436** (e.g., the locating feature **436** and the mounts **438**) are oriented along a depth D of the image capture device **400**, which facilitates insertion of the mounting member **800**, the heatsink **1200**, the tray **600**, etc., in a forward (front) direction (e.g., towards the window **412**), which is identified by arrow **1** (FIGS. **6**, **11**, **12**).

(76) In the particular embodiment illustrated, the front housing portion **402** includes: (first) mounts **438***i* that are configured to receive the mechanical fasteners **428** upon insertion through the apertures **422** in the flange **420**; (second) mounts **438***ii* that are configured to receive a plurality (series) of mechanical fasteners **440** (FIG. **11**) (e.g., screws, pins, rivets, etc.), which extend through the heatsink **1200** to thereby directly connect the heatsink **1200** to the front housing portion **402** in either a fixed or removable fashion; (third and fourth) mounts **438***iii*, **438***iv* that are configured to secure the tray **600** in relation to the housing portions **402**, **406**; and mounts **438***v* that are configured to receive a plurality (series) of mechanical fasteners **442** (FIG. **6**) (e.g., screws, pins, rivets, etc.), which extend through the interconnect mechanism **136** (FIGS. **1**B, **4**, **9**) such that the interconnect mechanism **136** is directly connected to the front housing portion **402** in either a fixed or removable fashion. Due to the mounting location and the configuration of the interconnect mechanism **136** and the image capture device **100**, the interconnect mechanism **136** not only facilitates connection of the image capture device **100** to an accessory, as indicated above, but provides a base (e.g., a platform, a stand, etc.) in the extended (open) position.

(77) To further facilitate assembly of the heatsink **1200** to the front housing portion **402**, it is envisioned that the heatsink **1200** may include one or more structural supports **1202** (FIGS. **11**, **12**) (e.g., braces **1204**) that are configured for connection to corresponding mounting locations within the internal compartment **408**. For example, it is envisioned that each structural support **1202** may include one or more openings **1206** that are configured to receive a corresponding mechanical fastener **1208** (e.g., screws, pins, rivets, etc.) to thereby mechanically connect the heatsink **1200** to the front housing portion **402**. Additionally, or alternatively, it is envisioned that the heatsink **1200** and the front housing portion **402** may be configured for connection (engagement) in an interference (press) fit arrangement, via an adhesive, or in any other suitable manner.

(78) The rear housing portion **406** supports various thermal and electrical components of the image

capture device **400** (e.g., circuitry and electronics supporting the operation and functionality thereof. For example, it is envisioned that the rear housing portion **406** may include one or more flexible printed circuits (FPCs), electrical connectors, one or more conductive elements to facilitate grounding of the rear housing portion **406**, etc.

(79) The rear housing portion **406** includes an additional, independent (second, rear) heatsink **444** (FIG. **7**), which is formed as a separate component of the image capture device **400** that is discrete from the heatsink **1200**. The incorporation of heatsink functionality into the rear housing portion **406** obviates the need for a separate heatsink that may otherwise be connected to the rear housing portion **406**, which simplifies assembly and construction of the image capture device **400** and contributes to a reduction in the overall form factor.

(80) To facilitate the conduction of heat (e.g., away from the ISLA **1000**, the PCB module **1300**, etc.), the rear housing portion **406** may include (e.g., may be formed partially or entirely from) any suitable material (e.g., aluminum) or combination of materials and may be thermally connected to one or more components of the image capture device **400** (e.g., to the image sensor(s) **312** (FIGS. **3**, **10**), the aforementioned image processor, the PCB module **1300**, etc.). It is envisioned that such thermal connections may be established in any suitable manner, such as, for example, by one or more graphite conductors. Integrating the heatsink **444** into the rear housing portion **406** allows for the elimination of any structural connection therebetween, thereby reducing the number of fasteners and connections required so as to simplify assembly of the image capture device **400**.

(81) Extending around its perimeter, the rear housing portion **406** defines a recess (groove) **446** (FIG. **8**) that is configured in correspondence with the tongue **432** (FIG. **6**) extending form the front housing portion **402**. The recess **446** is configured to receive the tongue **432** such that the housing portions **402**, **406** are connectable in a tongue-and-groove fashion. To enhance the connection between the housing portions **402**, **406** and further facilitate the formation of a watertight seal therebetween, it is envisioned that an adhesive **448** may be applied to the tongue **432** and/or the recess **446** prior to connection of the housing portions **402**, **406**.

(82) With reference now to FIGS. **13** and **14** as well, the tray **600** will be discussed. More specifically, FIG. **13** provides a vertical, cross-sectional view of the front housing portion **402** and the tray **600** upon installation, and FIG. **14** provides a vertical, cross-sectional view of the rear housing portion **406** and the tray **600** upon installation. The tray **600** supports the power source **700** and includes: one or more (front, first) flanges **602** (FIG. **11**); one or more (rear, second) flanges **604**; an opening **606**; and a plurality (series) of through-bores **608** (or other such openings). As described in further detail below, the tray **600** extends between (and engages) the housing portions **402**, **406** such that the tray **600** is secured therebetween.

(83) The flange(s) **602** interface with (connect to, extend into engagement with) the front housing portion **402**. More specifically, the flange(s) **602** are configured for insertion into the mounts **438***iii*, which include receptacles **450** that extend (vertically) upward from a base wall **452** of the front housing portion **402** within the internal compartment **408** and are recessed vertically into the tray **600** so as to define a tab **610** that is positioned between the receptacles **450**.

(84) In the particular embodiment illustrated, the tray **600** include a pair of flanges **602** and the front housing portion **402** includes a pair of mounts **438***ii* (e.g., a pair of receptacles **450**). It should be appreciated, however, that the particular number of flanges **602** and mounts **438***ii* may be altered in various embodiments without departing from the scope of the present disclosure. As such, embodiments of the image capture device **400** including both greater and fewer numbers of flanges **602** and mounts **438***ii* are also contemplated herein and would not be beyond the scope of the present disclosure.

(85) The flanges **604** interface with (connect to, extend into engagement with) the rear housing portion **406**. More specifically, the flanges **604** are configured for engagement (contact) with corresponding tabs **454** (FIG. **8**) that extend (project) from the rear housing portion **406** towards the front housing portion **402**. As seen in FIGS. **8** and **14**, each of the tabs **454** defines a chamfered

(beveled) surface **456** that is configured for engagement with a corresponding chamfered (beveled) surface **612** defined by each of the flanges **604** so as to guide and secure the flanges **604** beneath the tabs **454** during assembly of the housing portions **402**, **406**.
(86) In the particular embodiment illustrated, the tray **600** includes a pair of flanges **604** and the rear housing portion **406** includes four tabs **454**. It should be appreciated, however, that the particular number of flanges **604** and tabs **454** may be altered in various embodiments without departing from the scope of the present disclosure. As such, embodiments of the image capture device **400** including both greater and fewer numbers of flanges **604** and tabs **454** are also contemplated herein and would not be beyond the scope of the present disclosure.
(87) The opening **606** in the tray **600** is configured to receive a mechanical fastener **614** (e.g., a screw, pin, rivet, etc.) such that the mechanical fastener **614** extends through the tray **600** and into an opening **458** defined by the mount **438***iv* to further secure the tray **600** in relation to the front housing portion **402**.
(88) The through-bores **608** extend through the tray **600** and are configured to receive the mounts **438***v*, which extend inwardly from the front housing portion **402** and into the internal compartment **408**. As seen in FIG. **13**, upon assembly of the image capture device **400**, the mounts **438***v* extend into the through-bores **608**, which further contributes to reductions in the overall form factor of the image capture device **400**.
(89) The mounting member **800** is connected (secured) to the front housing portion **402** adjacent to the window **412** (FIGS. **4**, **11**) defined by the flange **420** and supports the cover **500** and the ISLA **1000**, each of which is configured for direct connection to the mounting member **800**, as discussed in further detail below. By directly connecting the ISLA **1000** to the mounting member **800**, physical, supportive connections between the ISLA **1000** and the heatsink **1200** can be reduced (if not entirely eliminated) so as to further simplify assembly of the image capture device **400**. The configuration of the mounting member **800** also eliminates any direct physical connection between the cover **500** and the heatsink **1200**, which also simplifies sealing of the image capture device **400**. More specifically, as described in further detail below, by sealing the mounting member **800** to the front housing portion **402** (via the sealing member **900**) and the ISLA **1000** (via the sealing member **1100**), sealing can be localized to the mounting member **800**, which allows the overall geometry and architecture of the image capture device **400** to be simplified by eliminating the need for a perimeter seal about the heatsink **1200**. Moreover, the seals established between the mounting member **800**, the front housing portion **402**, and the ISLA **1000** by the sealing members **900**, **1100** allow the image capture device **400** to remain watertight upon removal of the cover **500**.
(90) The mounting member **800** includes respective (front, first and rear, second) end portions **804**, **806** that are separated (delineated) by a (generally planar) base **808**. The end portions **804**, **806** respectively include (front, first and rear, second) collars **810**, **812** (FIGS. **11**, **12**) that extend from the base **808** in opposing (forward and rearward) directions. The base **808** is configured in correspondence with the flange **420** of the front housing portion **402** such that the base **808** seats within the window **412**, whereby, upon assembly of the image capture device **400**, the end portion **804** (e.g., the collar **810**) extends (is positioned) externally of the image capture device **400** and the end portion **806** (e.g., the collar **812**) extends into (is positioned internally within) the image capture device **400** (e.g., within the internal compartment **408**). The base **808** includes a (central) opening **814** that is configured to receive the ISLA **1000** and a plurality (series) of apertures **816** that extend therethrough. The apertures **816** are positioned in alignment (registration) with the apertures **422** (FIG. **11**) formed in the flange **420** and are configured to receive the mechanical fasteners **428** to thereby connect the mounting member **800** to the front housing portion **402**.
(91) The collar **810** (FIG. **11**) of the mounting member **800** is generally annular in configuration and extends outwardly from the base **808** in the forward direction **1**. The collar **810** includes a pair of ears **818** that are formed integrally (e.g., unitarily, monolithically) therewith and which extend radially (laterally) outward therefrom. The ears **818** are configured for releasable engagement with

the cover **500** such that the cover **500** is (repeatably) connectable to and disconnectable from the image capture device **400** via the mounting member **800**. While the collar **810** is illustrated as including a pair of diametrically opposed ears **818** in the illustrated embodiment, it should be appreciated that the particular number, location, and/or configuration of ears **818** may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the cover **500**).

(92) The collar **812** (FIG. **12**) is generally annular in configuration and extends outwardly from the base **808** in a rearward direction (e.g., toward the rear housing portion **406**), which is identified by the arrow **2** (FIGS. **6**, **11**, **12**). The collar **812** is spaced radially (laterally) inward from the apertures **816** extending through the base **808**. The collar **812** is configured for insertion into (through) the window **412** to further recess the mounting member **800** within the front housing portion **402**, thereby reducing the overall spatial requirements of the mounting member **800** and further contributing to reductions in the overall form factor of the image capture device **400**.

(93) The collar **812** includes an integral locating feature **820** (e.g., a detent **822** or other such projection) that is configured for receipt within a corresponding notch **1102** (FIG. **11**) defined by the sealing member **1100** to facilitate proper relative orientation of the mounting member **800** and the sealing member **1100**. Although shown as including a single locating feature **820** in the illustrated embodiment, it should be appreciated that the particular number and/or location of the locating feature(s) **820** may varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the sealing member **1100**).

(94) The collar **812** further includes a plurality (series) of eyelets **824** (FIG. **12**) extending radially (laterally) outward therefrom that are configured for reception within the reliefs **424** defined by the flange **420**, which facilitates proper orientation of the mounting member **800** and the front housing portion **402** and inhibits (if not entirely prevents) relative rotation therebetween. Each eyelet **824** defines an aperture **826** that extends partially through the mounting member **800**. The eyelets **824** and the apertures **826** are positioned in alignment (registration) with corresponding apertures **1002** (FIG. **12**) in the ISLA **1000** such that the ISLA **1000** is directly connectable to the mounting member **800** via a plurality (series) of mechanical fasteners **828** (FIG. **11**) (e.g., screws, pins, rivets, etc.) in either a fixed or removable fashion. Although shown as including three eyelets **824** and three apertures **826** that are spaced (approximately) equidistant from each other in the particular embodiment illustrated (e.g., such that the apertures **826** are separated by approximately 120°), it should be appreciated that the particular number and/or location of the eyelets **824** and the apertures **826** may varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **1000**).

(95) The sealing member **900** (FIGS. **11**, **12**) is located between the mounting member **800** and the front housing portion **402**. More specifically, the sealing member **900** is positioned about the collar **812** of the mounting member **800** within the channel **426** (FIG. **11**) defined by the flange **420** such that the sealing member **900** is positioned between the apertures **816** in the base **808** and the eyelets **824**, which are received by the reliefs **424** such that the eyelets **824** extend into the channel **426**. More specifically, the sealing member **900** is positioned radially (laterally) inward of the apertures **816** and laterally outward of the eyelets **824**. To facilitate such location, in certain embodiments, such as that illustrated, the sealing member **900** may include a (first) non-annular configuration that may be characterized as generally escutcheon or shield-shaped. Alternative configurations for the sealing member **900**, however, would not be beyond the scope of the present disclosure. For example, an embodiment of the sealing member **900** including a generally annular configuration is also contemplated herein. Additionally, while the sealing member **900** is illustrated as including a generally circular (annular) cross-sectional configuration, it should be appreciated that the particular cross-sectional configuration of the sealing member **900** may be varied in alternate embodiments without departing from the scope of the present disclosure. For example, it is envisioned that the sealing member **900** may instead include a (generally) elliptical or polygonal

cross-sectional configuration.
(96) The sealing member **900** defines a (central) opening **902** that is configured to receive the collar **812** of the mounting member **800** and the ISLA **1000**. The sealing member **900** is configured to form a watertight seal between the mounting member **800** and the front housing portion **402** and may include (e.g., may be formed partially or entirely from) any material or combination of materials suitable for this intended purpose. For example, in the illustrated embodiment, the sealing member **900** includes (e.g., is formed partially or entirely from) silicone. It should be appreciated, however, that the use of alternative material(s) of construction would not be beyond the scope of the present disclosure.
(97) With particular reference to FIGS. **10-12**, the ISLA **1000** will be discussed. The ISLA **1000** receives and focuses light and converts captured content into an electronic image signal that is processed to form an image. The ISLA **1000** includes a body **1004** supporting one or more image sensors (e.g., the image sensor(s) **312** (FIG. **3**) and one or more lenses (e.g., the lens(es) **104**), which receive and direct light onto the image sensor(s) **312**.
(98) The ISLA **1000** extends through an opening **1210** (FIG. **12**) in the heatsink **1200**, which allows for direct connection (securement) of the ISLA **1000** to the mounting member **800** (e.g., the rear end portion **806**) in the forward direction **1** from within the image capture device **400** (e.g., from within the internal compartment **408**), whereby the ISLA **1000** is operatively (indirectly) connected to the front housing portion **402** via the mounting member **800**. Upon assembly of the image capture device **400**, the ISLA **1000** (e.g., the lens **104**) extends through the window **412** defined by the flange **420** of the front housing portion **402**, as seen in FIG. **4**. To facilitate such connection, the ISLA **1000** includes a collar **1006** supporting a plurality (series) of wings **1008** that extend radially outward therefrom, which include the aforedescribed apertures **1002**. The collar **1006** is concentrically positioned so as to define a (generally annular) shoulder **1010** that supports the sealing member **1100**. Upon assembly of the image capture device **400**, the mechanical fasteners **828** are inserted through the apertures **1002** in the wings **1008** (in the forward direction **1**) and into the apertures **826** (FIG. **12**) in the eyelets **824** on the collar **812** of the mounting member **800**, thereby directly securing together the ISLA **1000** and the mounting member **800**.
(99) Although shown as including three wings **1008** that are spaced (approximately) equidistant from each other in the particular embodiment illustrated (e.g., such that the wings **1008** are separated by approximately 120°), it should be appreciated that the particular number and/or location of the wings **1008** may varied without departing from the scope of the present disclosure (e.g., depending upon spatial requirements, the particular configurations of the ISLA **1000** and the heatsink **1200**, etc.).
(100) In certain embodiments, such as that illustrated throughout the figures, it is envisioned that the heatsink **1200** may include one or more recesses **1212** (FIG. **11**) that are configured in correspondence with the sealing member **1100** (as described in further detail below) and a plurality (series) of reliefs **1214** (e.g., cutouts, notches, indentations, etc.) that are configured in correspondence with the wings **1008** of the ISLA **1000**. As such, in the illustrated embodiment, the heatsink **1200** includes three reliefs **1214** that are spaced (approximately) equidistant from each other (e.g., such that the reliefs **1214** are separated by approximately) 120°. It should be appreciated, however, that the particular number and/or location of the reliefs **1214** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular number and/or location of the wings **1008** included on the ISLA **1000**).
(101) Upon assembly of the image capture device **400**, the ISLA **1000** extends through the opening **1210** and the wings **1008** extend through the reliefs **1214**, which not only facilitates proper alignment (registration) of the ISLA **1000** and the heatsink **1200**, but reduces the overall spatial requirements of the heatsink **1200** and the ISLA **1000**, thereby further contributing to reductions in the overall form factor of the image capture device **400**.
(102) By directly connecting the ISLA **1000** to the mounting member **800**, physical, supportive

connections between the ISLA **1000** and the heatsink **1200** can be reduced, thereby simplifying assembly of the image capture device **400**. More specifically, the architecture of the front housing portion **402** (e.g., the bucket **404**) described herein allows for a complete elimination of any physical connection between the ISLA **1000** and the heatsink **1200**. Additionally, connection of the ISLA **1000** directly to the mounting member **800**, rather than via an intermediate component (such as the heatsink **1200**, for example), enables precise alignment of the ISLA **1000** and the mounting member **800**, which, thus, enables precise alignment of the cover **500**, the mounting member **800**, and the ISLA **1000**. By increasing precision in the alignment of the cover **500** and the ISLA **1000** (via positioning of the mounting member **800** located therebetween), basic functionality of the image capture device **400** can be improved. Moreover, direct connection of the ISLA **1000** to the mounting member **800** increases precision in the relative alignment of the ISLA **1000** and the front housing portion **402**, which not only simplifies assembly of the image capture device **400** by reducing the number of alignments that must occur, but enhances the overall fit and finish of the image capture device **400**.

(103) The sealing member **1100** is located between the mounting member **800** and the ISLA **1000**. More specifically, the sealing member **1100** is positioned about the ISLA **1000** such that the sealing member **1100** is supported by the shoulder **1010** (FIG. **11**) defined by the collar **1006**. The sealing member **1100** forms watertight seals with both the mounting member **800** and the ISLA **1000** upon assembly of the image capture device **400**, thereby obviating any need to form a direct seal between the ISLA **1000** and the front housing portion **402**, and may include any material(s) suitable for this intended purpose. For example, in the particular embodiment illustrated throughout the figures, the sealing member **1100** includes (e.g., is formed partially or entirely from) conductive silicone, which facilitates grounding of the heatsink **1200** relative to the mounting member **800** (via the sealing member **1100**). It should be appreciated, however, that the use of alternative material(s) of construction would not be beyond the scope of the present disclosure.

(104) The sealing member **1100** includes a (second, generally annular) configuration that differs from that of the sealing member **900**. More specifically, the sealing member **1100** includes a rib **1104** and a plurality (series) of feet **1106** that extend radially outward from the rib **1104**. The rib **1104** is (generally) annular in configuration and defines a (central) opening **1108** for receipt of the ISLA **1000**, an inner transverse cross-sectional dimension (e.g., an inner diameter) that exceeds an outer transverse cross-sectional dimension (e.g., an outer diameter) defined by the collar **1006** of the ISLA **1000** and an outer transverse cross-sectional dimension (e.g., an outer diameter) that is less than an inner transverse cross-sectional dimension (e.g., an inner diameter) defined by the collar **812** of the mounting member **800**, which allows the rib **1104** to be positioned concentrically about the collar **1006** and concentrically within the collar **812** upon assembly of the image capture device **400**. The relative dimensioning between the collar **1006**, the rib **1104**, and the collar **812** allows the sealing member **1100** to simultaneously contact (engage), and thereby form watertight seals with, the ISLA **1000** and the mounting member **800**, thus, sealing together the ISLA **1000**, the mounting member **800**, and the front housing portion **402**. More specifically, the sealing member **1100** seals with the ISLA **1000** at the interface of the rib **1104** and the collar **1006** and with the mounting member **800** at the interfaces between the collar **812**, the rib **1104**, and outer (front, forward) surfaces **1110** (FIG. **11**) of the feet **1106**.

(105) The feet **1106** are configured for reception by and seating within the recesses **1212** defined by the heatsink **1200** and are circumferentially spaced from each other so as to define windows **1112** therebetween. The windows **1112** are positioned in alignment (registration) with the wings **1008** extending from the collar **1006** of the ISLA **1000** such that the wings **1008** and, thus, the mechanical fasteners **828**, are received by (and extend through) the windows **1112** upon assembly of the image capture device **400**. While the illustrated embodiment of the sealing member **1100** is shown as including three feet **1106** that are spaced (approximately) equidistant from each other (e.g., such that centerpoints thereof are separated by approximately 120°) and three windows **1112**

that are spaced (approximately) equidistant from each other (e.g., such that centerpoints thereof are separated by approximately 120°), it should be appreciated that the specific number, location, and/or configuration of the feet **1106** and/or the windows **1112** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **1000**).

(106) In certain embodiments, such as that illustrated, it is envisioned that the sealing member **1100** may include one or more contacts **1114** (FIGS. **10**, **12**) to facilitate grounding of the heatsink **1200** relative to the mounting member **800**. Although shown as being positioned on rear surfaces **1116** (FIG. **10**) of the feet **1106**, it should be appreciated that the particular location of the contact(s) **1114** may be varied in alternate embodiments without departing from the scope of the present disclosure.

(107) As indicated above, the locating feature **820** (FIG. **12**) extending from the collar **812** of the mounting member **800** is configured for insertion into the aforementioned notch **1102**, which is provided in (defined by) one of the feet **1106** on the sealing member **1100**, to facilitate proper relative orientation of the mounting member **800** and the sealing member **1100**. Although shown as including a single notch **1102** in the illustrated embodiment, it should be appreciated that the particular number and/or location of the notch(es) **1102** may varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the mounting member **800**). For example, it is envisioned that each foot **1106** may define a notch **1102** that is configured to receive a corresponding locating feature **820** extending from the collar **812**.

(108) With reference now to FIGS. **4**, **11**, and **12**, the cover **500** will be discussed. The cover **500** functions as a removable cap that protects and conceals the ISLA **1000**. Although shown as being generally square-shaped in configuration throughout the figures, it should be appreciated that the specific configuration of the cover **500** may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **1000** and/or the front housing portion **402**, the desired aesthetic appearance of the image capture device **400**, etc.).

(109) The cover **500** and the mounting member **800** are configured for direct, releasable connection. More specifically, the cover **500** is configured for removable connection to the front end portion **804** of the mounting member **800** via the ears **818**, which facilitates (repeated) connection and disconnection of the cover **500**, as indicated above. For example, the cover **500** may include one or more pockets (or other such recesses or openings) that are configured in correspondence with the ears **818**. To facilitate connection and disconnection of the cover **500**, it is envisioned that the ears **818** may include (or otherwise define) angled (e.g., chamfered) bearing surfaces **830** (FIG. **4**) that are configured to urge the cover **500** outwardly (away from the front housing portion **402**) upon rotation of the cover **500**.

(110) Directly connecting the cover **500** to the mounting member **800** allows for the elimination of any interposed components (e.g., the heatsink **1200**) between the cover **500**, the mounting member **800**, and the front housing portion **402**. The elimination of interposed components between the cover **500**, the mounting member **800**, and the front housing portion **402** reduces the number of components that must be properly aligned during assembly of the image capture device **400**, which not only simplifies the assembly process, but improves the overall alignment of the cover **500** relative to the front housing portion **402** and the ISLA **1000**, thereby enhancing the overall fit and finish of the image capture device **400**.

(111) To further enhance sealing of the image capture device **400**, it is envisioned that the image capture device **400** may include a variety of supplemental sealing members **1500**, which may include (e.g., may be formed from) any suitable material or combination of materials, such as a foam, silicone, one or more polymeric materials, a pressure-sensitive adhesive, a heat-activated film, etc. For example, FIGS. **15** and **16** provide front and rear perspective views of the image capture device **400**, respectively, and illustrate a sealing member **1500***i* that is configured to form a

watertight seal between the front housing portion **402** and the door **114** (FIG. **1**A); a sealing member **1500***ii* that is configured to form a watertight seal about the light pipe **418**; a sealing member **1500***iii* that is configured to form a watertight seal about the one or more upper (top) microphones; a sealing member **1500***iv* that is configured to form a watertight seal about one or more front microphones; a sealing member **15000***v* that is configured to form a watertight seal about one or more rear microphones; a sealing member **1500***vi* that is configured to form a watertight seal about one or more vents that allow for pressure equalization with the external environment by allowing air to pass therethrough; a sealing member **1500***vii* that is configured to form a watertight seal about one or more speakers; and a sealing member **1500***viii* that is configured to form a watertight seal about the display **416**.

(112) With reference now to FIGS. **17-23**, the PCB module **1300** and the chassis **1400** will be discussed. More specifically, FIG. **17** provides a rear, perspective view of the PCB module **1300** and the chassis **1400** upon assembly of the image capture device **400**; FIGS. **18** and **19** provide rear, perspective views of the PCB module **1300** and the chassis **1400** shown removed from the image capture device **400** and connected to the heatsink **1200**; FIG. **20** provides a side, plan view of the PCB module **1300** and the chassis **1400** shown removed from the image capture device **400** and connected to the heatsink **1200**; FIG. **21** provides a top, plan view of the PCB module **1300** and the chassis **1400** shown removed from the image capture device **400** and connected to the heatsink **1200**; FIG. **22** provides a rear, perspective view of the PCB module **1300** shown in a first configuration; and FIG. **23** provides a rear, perspective view of the PCB module **1300** shown in a second configuration.

(113) The PCB module **1300** includes: a front (first) PCB assembly **1302**, which defines respective front and rear faces **1304**, **1306** (FIGS. **22**, **23**); a rear (second) PCB assembly **1308**, which defines respective front and rear faces **1310**, **1312** (FIGS. **22**, **23**); and a connector **1314**, which extends between the PCB assemblies **1302**, **1308** and includes opposing (first and second) end portions **1316**, **1318** (FIGS. **22**, **23**) that are connected by a body portion **1320**. The PCB module **1300** is connected to the heatsink **1200** and the chassis **1400** via a plurality (series) of mechanical fasteners **1322** (e.g., screws, pins, rivets, etc.) in either a fixed or removable fashion, as described in further detail below, whereby PCB module **1300** is operatively (indirectly) connected to the front housing portion **402** via the heatsink **1200**. The PCB module **1300** (e.g., the PCB assemblies **1302**, **1308**) supports various electrical and/or thermal components of the image capture device **400**, including at least one (one or more) higher-power (e.g., heat-generating, first) components **1324** (e.g., an SOC **1326** for the image capture device **400**, one or more (at least one) power management integrated circuits (PMICs) **1328**, and the image sensor **312** (FIG. **10**), which is included on the ISLA **1000**) and at least one (one or more) lower-power (second) components **1330**, which are distributed across the PCB assemblies **1302**, **1308**, the ISLA **1000**, etc. More specifically, in the particular embodiment illustrated, the higher-power components **1324** include (first and second) PMICs **1328***i*, **1328***ii* (FIGS. **9**, **17**) and the lower-power components **1330** include an integrated circuit (IC) **1332** (FIGS. **20**, **22**, **23**) (e.g., a WiFi IC, an embedded multi-media card, etc.) and an SD card reader (socket) **1333**. It should be appreciated, however, that the particular number and/or functionality of the PMICs **1328** and/or the ICs **1332** may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending upon the particular capabilities of the image capture device **400**). Additionally, it should be appreciated that partitioning (distribution) of the various electrical and/or thermal components throughout the image capture device **400** may be dependent upon a variety of factors, including, for example, the area available on different sections of the PCB assemblies **1302**, **1308**.

(114) The front and rear PCB assemblies **1302**, **1308** include rigid bodies **1334**, **1336** (FIGS. **22**, **23**), respectively, which support the higher-power components **1324** and the lower-power components **1330**. The bodies **1334**, **1336** include a laminated construction that is defined by layers (panels) **1338** of material that are arranged in adjacent relation. The layers **1338** may include (e.g.,

may be formed from) any suitable material or combination of materials such as, for example, one or more of a (glass-reinforced) epoxy sheet, PTFE, one or more metallic materials, etc., thereby attributing the requisite rigidity to the respective bodies **1334**, **1336** of the front and rear PCB assemblies **1302**, **1308**. While the bodies **1334**, **1336** are each illustrated as including a pair of layers **1338** (e.g., an inner layer **1338***i* and an outer layer **1338***o*) in the particular embodiment illustrated, it should be appreciated that the particular number of layers **1338** may be altered in various embodiments without departing from the scope of the present disclosure (e.g., to increase or decrease the rigidity of the bodies **1334**, **1336**). As such, embodiments of the bodies **1334**, **1336** including both greater and fewer numbers of layers **1338** are also contemplated herein and would not be beyond the scope of the present disclosure.
(115) In contrast to the rigid construction of the PCB assemblies **1302**, **1308**, the connector **1314** includes a flexible (pliable, elastic), non-rigid construction, which facilitates reconfiguration of the PCB module **1300** between the first configuration (FIG. **22**) and the second configuration (FIG. **23**). As such, throughout the present disclosure, the connector **1314** may also be referred to as the "flexible connector." It is envisioned that the connector **1314** may include (e.g., may be formed from) any material or combination of materials suitable for the intended purpose of facilitating deformation (bending) in the manner described herein, including, for example, one or more polyimide or polyester films.
(116) As seen in FIG. **22**, when the PCB module **1300** is in the first configuration, the PCB assemblies **1302**, **1308** are oriented in non-parallel relation. By contrast, when the PCB module **1300** is in the second configuration, the PCB assemblies **1302**, **1308** are oriented in generally parallel relation, as seen in FIG. **23**. More specifically, in the first configuration, the PCB module **1300** includes a generally L-shaped configuration in which the PCB assemblies **1302**, **1308** subtend an angle α (FIG. **22**) therebetween that lies substantially within the range of (approximately) 75 degrees to (approximately) 105 degrees (e.g., (approximately) 90 degrees), whereby the body portion **1320** of the connector **1314** and the rear PCB assembly **1308** are oriented in generally parallel relation and the front PCB assembly **1302** extends in generally orthogonal relation to both the body portion **1320** of the connector **1314** and the rear PCB assembly **1308**. In the second configuration, the PCB module **1300** includes a generally U-shaped configuration in which the angle α subtended therebetween lies substantially within the range of (approximately) 165 degrees to (approximately) 195 degrees (e.g., (approximately) 180 degrees), whereby the body portion **1320** of the connector **1314** is oriented in generally orthogonal relation to each of the PCB assemblies **1302**, **1308** such that the PCB assemblies **1302**, **1308** collectively define a (generally U-shaped) chamber **1340** that is configured to receive the chassis **1400**, as described in further detail below.
(117) The generally U-shaped configuration of the PCB module **1300** realized upon assembly of the image capture device **400** offers several advantages. For example, the generally U-shaped configuration allows the various electrical and/or thermal components of the image capture device **400** to be included on a single PCB module **1300**, which allows for a more compact internal design and further contributes to reductions in the overall form factor of the image capture device **400**. Additionally, the generally U-shaped configuration facilitates and simplifies assembly by improving visibility (e.g., via improved access to the component(s) housed (accommodated) within the chamber **1340**) and reducing (if not entirely eliminating) blind steps.
(118) As seen in FIGS. **22** and **23**, for example, the opposing end portions **1316**, **1318** of the connector **1314** are respectively connected to the PCB assemblies **1302**, **1308**, which facilitates reconfiguration of the PCB module **1300** between the first and second configurations. In the particular embodiment illustrated, the opposing end portions **1316**, **1318** extend into, and are positioned within, the rigid bodies **1334**, **1336**, respectively. More specifically, the layers **1338***i*, **1338***o* of the rigid body **1334** are positioned about the end portion **1316**, and the layers **1338***i*, **1338***o* of the rigid body **1336** are positioned about the end portion **1318**, whereby the end portions **1316**, **1318** are sandwiched between the layers **1338** of the rigid bodies **1334**, **1336**, respectively. It

should be appreciated, however, that the PCB assemblies **1302**, **1308** and the connector **1314** may be secured together in any suitable manner including, for example, via one or more mechanical fasteners (e.g., screws, pins, rivets, etc.), via an adhesive, via ultrasonic welding, etc.

(119) Depending upon the particular electrical and thermal requirements of the image capture device **400**, it is envisioned that the various electrical and/or thermal components thereof may be arranged in a variety of orientations. For example, in the particular embodiment illustrated, the higher-power component(s) **1324** are located on (secured to) the rear PCB assembly **1308**, which allows the higher-power component(s) **1324** to interface with the rear housing portion **406** (FIG. **7**) and, thus, the heatsink **444**, whereas the lower-power component(s) **1330** are located on (secured to) the front PCB assembly **1302**. In such embodiments, in order to support the conduction of heat away from the PCB module **1300**, it is envisioned that the rear PCB assembly **1308** may be thermally connected to the rear housing portion **406** (e.g., the heatsink **444**) via one or more suitable thermal connectors **1342** (FIG. **12**) (e.g., graphite sheets **1344** or other such thermally-conductive members) so as to facilitate the transfer of thermal energy from the rear PCB assembly **1308** to the heatsink **444**.

(120) In another embodiment of the disclosure, however, it is envisioned that the higher-power component(s) **1324** may be located externally of the chamber **1340** and that the lower-power component(s) **1330** may be located within the chamber **1340**. More specifically, the higher-power component(s) **1324** may be secured to the front face **1304** of the front PCB assembly **1302** and the rear face **1312** of the rear PCB assembly **1308**, which allows for a thermal interface between the higher-power component(s) **1324** and the heatsinks **444**, **1200**, and the lower-power component(s) **1330** may be secured to the rear face **1306** of the front PCB assembly **1302** and the front face **1310** of the rear PCB assembly **1308**. In such embodiments, in order to support the conduction of heat away from the PCB module **1300**, it is envisioned that the PCB assemblies **1302**, **1308** may be thermally connected to the heatsinks **444**, **1200** via one or more of the aforementioned thermal connectors **1342**.

(121) With reference to FIGS. **17-21** in particular, the chassis **1400** will be discussed. The chassis **1400** is configured for positioning (reception) within the chamber **1340** defined by the PCB module **1300** (in the second configuration), whereby, upon assembly of the image capture device **400**, the chassis **1400** is positioned between the PCB assemblies **1302**, **1308**. The chassis **1400** includes a (metallic) frame **1402** (FIGS. **19**, **20**) that provides a bridge between the PCB assemblies **1302**, **1308** and structurally supports the PCB module **1300** within the front housing portion **402** of the image capture device **400**. The architecture and configuration of the chassis **1400** allows the PCB module **1300** to be structurally supported solely by the front housing portion **402**, which eliminates any need for structural connections between the PCB module **1300** and the rear housing portion **406**.

(122) As seen in FIGS. **18** and **19**, the PCB module **1300** is mechanically connected to both the heatsink **1200** and the chassis **1400** via the aforementioned mechanical fasteners **1322**, which include: one or more mechanical fasteners **1322***a*; one or more mechanical fasteners **1322***b*; one or more mechanical fasteners **1322***c*; and one or more mechanical fasteners **1322***d*. More specifically, the mechanical fastener(s) **1322***a* extend through the rear PCB assembly **1308** and into the heatsink **1200**, the mechanical fastener(s) **1322***b* extend through the rear PCB assembly **1308** and into the chassis **1400**, the mechanical fastener(s) **1322***c* extend through the front PCB assembly **1302** and into the heatsink **1200**, and the mechanical fastener(s) **1322***d* extend through the PCB assemblies **1302**, **1308**, through the chassis **1400**, and into the heatsink **1200**.

(123) In certain embodiments of the disclosure, it is envisioned that the chassis **1400** may include (or otherwise support) one or more electrical connectors **1404**. For example, in the particular embodiment illustrated, the chassis **1400** includes a single USB connector **1406**. It should be appreciated, however, that the particular number and/or configuration of the electrical connector(s) **1404** may be varied in alternate embodiments without departing from the scope of the present

disclosure.
(124) With general reference now to FIGS. **4**, **6**, **8**, **9**, **11-14**, **17-23**, assembly of the image capture device **400** will be discussed. Initially, the mounting member **800** (FIG. **1**) is (externally) connected to the front housing portion **402** via the mechanical fasteners **428** (FIG. **11**), which are inserted into the mounts **438***i* (FIG. **6**) in the rearward direction which is identified by the arrow **2**. More specifically, the base **808** is positioned adjacent to the flange **420** such that the sealing member **900** (FIGS. **11**, **12**) is positioned and compressed therebetween so as to form a watertight seal between the front housing portion **402** and the mounting member **800**. Thereafter, the heatsink **1200** is connected (secured) to the front housing portion **402** via insertion of the mechanical fasteners **440** through the heatsink **1200** and into the mounts **438***ii* (FIG. **6**), and any necessary electrical connections are made (e.g., between the main board, the power source **700**, etc.).
(125) With the heatsink **1200** in place, the sealing member **1100** is brought into contact (engagement) with the mounting member **800** by positioning the sealing member **1100** such that the rib **1104** (FIG. **11**) is received by the collar **812**. Thereafter, the ISLA **1000** is positioned such that the ISLA **1000** extends through the heatsink **1200** (via the opening **1210**), through the sealing member **1100**, through the front housing portion **402** (via the window **412**), through the sealing member **900**, and through the mounting member **800**. When so positioned, the collar **1006** of the ISLA **1000** is received by the rib **1104** of the sealing member **1100** and the wings **1008** of the ISLA **1000** are received by the windows **1112** defined between the feet **1106** of the sealing member **1100**. The ISLA **1000** is then connected to the mounting member **800** via forward insertion of the mechanical fasteners **828** through the apertures **1002** (FIG. **12**) in the wings **1008**, through the sealing member **1100**, and into the apertures **826** in the eyelets **824** on the mounting member **800**, thereby compressing the sealing member **1100** between the ISLA **1000** and the mounting member **800** such that the ISLA **1000**, the mounting member **800**, and the front housing portion **402** are sealed together via the sealing members **900**, **1100**.
(126) Either prior or subsequent to installation of the ISLA **1000**, the tray **600** (FIGS. **13**, **14**) is installed and connected to the front housing portion **402**. More specifically, the front flanges **602** are inserted into the mounts **438***iii* (FIG. **6**) (e.g., the receptacles **450**) and the mounts **438***v* on the front housing portion **402** are inserted into the through-bores **608**. Thereafter, the mechanical fastener **614** is inserted through the opening **606** in the tray **600** and into the mount **438***iv* (FIG. **6**) in the front housing portion **402** to thereby secure the tray **600** in relation to the front housing portion **402**. The power source **700** can then be installed and positioned as illustrated in FIG. **9** such that the power source **700** is supported by the tray **600**.
(127) Following connection of the ISLA **1000** to the mounting member **800**, the PCB module **1300** (FIGS. **9**, **17-23**) is installed via direct connection to the heatsink **1200** and, thus, operative (indirect) connection to the front housing portion **402**. More specifically, with the PCB module **1300** in the first configuration (FIG. **22**), the PCB module **1300** is inserted into the internal compartment **408** (FIG. **6**) defined by the front housing portion **402** such that the front PCB assembly **1302** is oriented in generally parallel relation to the heatsink **1200**. The front PCB assembly **1302** is then connected to the heatsink **1200** via the mechanical fastener(s) **1322***c*. Following connection of the front PCB assembly **1302** to the heatsink **1200**, the rear PCB assembly **1308** and the chassis **1400** are connected via the mechanical fastener(s) **1322***b*, and the PCB module **1300** is reconfigured into the second configuration (FIGS. **17-21**, **23**), during which, the connector **1314** deforms (e.g., bends) in the manner reflected in the transition between FIGS. **22** and **23**. With the PCB module **1300** in the second configuration, the chassis **1400** housed within the chamber **1340** between the PCB assemblies **1302**, **1308**, and the rear PCB assembly **1308** is connected to the heatsink **1200** via the mechanical fastener(s) **1322***a* and to the chassis **1400** and the front PCB assembly **1302** via the mechanical fastener(s) **1322***c*.
(128) As indicated above, the various electrical and/or thermal components of the image capture device **400** may be arranged in a variety of orientations (e.g., depending upon the particular

electrical and thermal requirements of the image capture device **400**). As such, it is envisioned that reconfiguration of the PCB module **1300** into the second configuration may result in positioning of the lower-power component(s) **1330** (FIGS. **20**, **22**, **23**) adjacent to the heatsink **1200** and positioning of the higher-power component(s) **1324** adjacent to the heatsink **444**. Alternatively, it is envisioned that reconfiguration of the PCB module **1300** into the second configuration may result in positioning of the higher-power component(s) **1324** such that they are located externally of the chamber **1340** and directed outwardly (e.g., towards the heatsinks **444**, **1200**) and positioning of the lower-power component(s) **1330** within the chamber **1340** such that they are directed inwardly (e.g., away from the heatsinks **444**, **1200**).

(129) Following installation of the PCB module **1300**, the PCB module **1300** is thermally and/or electrically connected to heatsink **1200** and/or the heatsink **444** (FIG. **7**) on the rear housing portion **406**. More specifically, depending upon the particular configuration of the image capture device **400** and the PCB module **1300**, it is envisioned that the higher-power component(s) **1324** (e.g., the camera SOC **1326** and the PMIC(s) **1328**) may be thermally connected to the heatsink **444** (e.g., exclusively) or, alternatively, that the higher-power component(s) **1324** may be thermally connected to the heatsink **444** and the heatsink **1200** so as to divide the thermal load generated during operation of the image capture device **400** between the heatsinks **444**, **1200**.

(130) Thereafter, the adhesive **448** (FIG. **8**) is applied to the recess **446** defined by the rear housing portion **406** and/or the tongue **432** extending from the front housing portion **402**. Before the adhesive **448** dries (cures), the respective front and rear housing portions **402**, **406** are connected via insertion of the tongue **432** into the recess **446**, during which, the tabs **454** on the rear housing portion **406** engage (contact) the rear flanges **604** on the tray **600**. The adhesive **448** is then allowed to dry (cure) to connect (secure) together the housing portions **402**, **406** and form a seal therebetween, establishing a watertight internal environment for the components of the image capture device **400**. The interconnect mechanism **136** (FIGS. **1**B, **4**, **9**) is then connected to the front housing portion **402** via insertion of the mechanical fasteners **442** (FIG. **6**) through the interconnect mechanism **136** and into the mounts **438***v*, and the cover **500** is connected to the mounting member **800** via the ears **818** (FIG. **1**) on the collar **810**.

(131) While the present disclosure has been described in connection with certain embodiments, it is to be understood that the present disclosure is not to be limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

(132) Persons skilled in the art will understand that the various embodiments of the present disclosure and shown in the accompanying figures constitute non-limiting examples, and that additional components and features may be added to any of the embodiments discussed hereinabove without departing from the scope of the present disclosure. Additionally, persons skilled in the art will understand that the elements and features shown or described in connection with one embodiment may be combined with those of another embodiment without departing from the scope of the present disclosure to achieve any desired result and will appreciate further features and advantages of the presently disclosed subject matter based on the description provided. Variations, combinations, and/or modifications to any of the embodiments and/or features of the embodiments described herein that are within the abilities of a person having ordinary skill in the art are also within the scope of the present disclosure, as are alternative embodiments that may result from combining, integrating, and/or omitting features from any of the disclosed embodiments.

(133) Use of the term “optionally” with respect to any element of a claim means that the element may be included or omitted, with both alternatives being within the scope of the claim. Additionally, use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially

of," and "comprised substantially of." Accordingly, the scope of protection is not limited by the description set out above, but is defined by the claims that follow, and includes all equivalents of the subject matter of the claims.
(134) In the preceding description, reference may be made to the spatial relationship between the various structures illustrated in the accompanying drawings, and to the spatial orientation of the structures. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the structures described herein may be positioned and oriented in any manner suitable for their intended purpose. Thus, the use of terms such as "above," "below," "upper," "lower," "inner," "outer," "left," "right," "upward," "downward," "inward," "outward," "horizontal," "vertical," etc., should be understood to describe a relative relationship between the structures and/or a spatial orientation of the structures. Those skilled in the art will also recognize that the use of such terms may be provided in the context of the illustrations provided by the corresponding figure(s).
(135) Additionally, terms such as "generally," "approximately," "substantially," and the like should be understood to include the numerical range, concept, or base term with which they are associated as well as variations in the numerical range, concept, or base term on the order of 25% (e.g., to allow for manufacturing tolerances and/or deviations in design). For example, the term "generally parallel" should be understood as referring to an arrangement in which the pertinent components (structures, elements) subtend an angle therebetween that is equal to 180° as well as an arrangement in which the pertinent components (structures, elements) subtend an angle therebetween that is greater than or less than 180° (e.g., ±25%). The term "generally parallel" should thus be understood as encompassing configurations in which the pertinent components are arranged in parallel relation.
(136) Although terms such as "first," "second," "third," etc., may be used herein to describe various operations, elements, components, regions, and/or sections, these operations, elements, components, regions, and/or sections should not be limited by the use of these terms in that these terms are used to distinguish one operation, element, component, region, or section from another. Thus, unless expressly stated otherwise, a first operation, element, component, region, or section could be termed a second operation, element, component, region, or section without departing from the scope of the present disclosure.
(137) Each and every claim is incorporated as further disclosure into the specification and represents embodiments of the present disclosure. Also, the phrases "at least one of A, B, and C" and "A and/or B and/or C" should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

Claims

1. An image capture device comprising: a front housing portion defining an internal compartment; a rear housing portion connected to the front housing portion so as to form a watertight seal therebetween; a tray positioned between the front housing portion and the rear housing portion; a power source supported by the tray; a heatsink connected to the front housing portion and configured to distribute thermal energy through the image capture device; an integrated sensor-lens assembly (ISLA) extending through the heatsink, the ISLA including: an image sensor; and a lens positioned to receive and direct light onto the image sensor; a mounting member connected to the front housing portion and to the ISLA such that the ISLA is indirectly connected to the front housing portion via the mounting member; a printed circuit board (PCB) module connected to the heatsink, the PCB module including: a front PCB assembly and a rear PCB assembly each supporting one or more electrical and/or thermal components of the image capture device; and a flexible connector extending between the front PCB assembly and the rear PCB assembly, the flexible connector facilitating reconfiguration of the PCB module during assembly of the image

capture device between a first configuration, in which the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation, and a second configuration, in which the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation; and a chassis positioned between the front PCB assembly and the rear PCB assembly, the chassis including a frame structurally supporting the PCB module.

2. The image capture device of claim 1, wherein the front housing portion defines a window configured to receive the mounting member such that the mounting member extends through the window and into the internal compartment.

3. The image capture device of claim 2, wherein the mounting member includes a front end portion extending externally of the image capture device and a rear end portion extending into the image capture device, the ISLA connected to the rear end portion of the mounting member.

4. The image capture device of claim 3, wherein the front end portion of the mounting member is configured for releasable connection to a cover for the image capture device such that the cover is repeatedly connectable to and disconnectable from the image capture device via the mounting member.

5. The image capture device of claim 1, wherein the tray defines through-bores configured to receive mounts extending inwardly from the front housing portion, the mounts configured to receive fasteners that connect the front housing portion to an interconnect mechanism configured for engagement with an accessory such that the image capture device is connectable to the accessory via the interconnect mechanism.

6. The image capture device of claim 5, wherein the tray includes: a front flange engaging the front housing portion; and a rear flange engaging the rear housing portion.

7. The image capture device of claim 1, wherein the PCB module is generally L-shaped in the first configuration and generally U-shaped in the second configuration, the PCB module defining a generally U-shaped chamber in the second configuration that is configured to receive the chassis such that the chassis is positioned between the front PCB assembly and the rear PCB assembly.

8. The image capture device of claim 1, wherein the front PCB assembly and the rear PCB assembly each include a rigid body with a laminated construction defined by layers arranged in adjacent relation.

9. The image capture device of claim 8, wherein the flexible connector includes a first end portion and a second end portion, the first end portion positioned within the rigid body of the front PCB assembly between the layers thereof and the second end portion positioned within the rigid body of the rear PCB assembly between the layers thereof.

10. The image capture device of claim 1, wherein the chassis includes at least one electrical connector.

11. A method of assembling an image capture device, the method comprising: connecting a first heatsink to a front housing portion of the image capture device, wherein the first heatsink includes an enclosed opening extending therethrough; inserting an integrated sensor-lens assembly (ISLA) through the enclosed opening in the first heatsink, wherein the ISLA includes: an image sensor; and a lens positioned to receive and direct light onto the image sensor; inserting a printed circuit board (PCB) module including a front PCB assembly and a rear PCB assembly into the front housing portion with the PCB module in a first configuration in which the front PCB assembly and the rear PCB assembly are oriented in non-parallel relation; reconfiguring the PCB module from the first configuration into a second configuration in which the front PCB assembly and the rear PCB assembly are oriented in generally parallel relation; connecting the PCB module to the first heatsink; and thermally connecting the PCB module to a rear housing portion including a second heatsink, wherein the first heatsink and the second heatsink are formed as discrete components of the image capture device, wherein reconfiguring the PCB module from the first configuration into the second configuration includes orienting at least one first component on the PCB module such that that the at least one first component is directed outwardly towards the first heatsink and the

second heatsink and orienting at least one second component on the PCB module such that the at least one second component is directed inwardly away from the first heatsink and the second heatsink, the at least one first component including a system-on-chip and at least one power management integrated circuit, and the at least one second component including at least one integrated circuit.

12. The method of claim 11, wherein thermally connecting the PCB module to the rear housing portion includes thermally connecting the second heatsink to a system-on-chip and at least one power management integrated circuit.

13. The method of claim 11, further comprising thermally connecting the at least one first component to the first heatsink and the second heatsink.